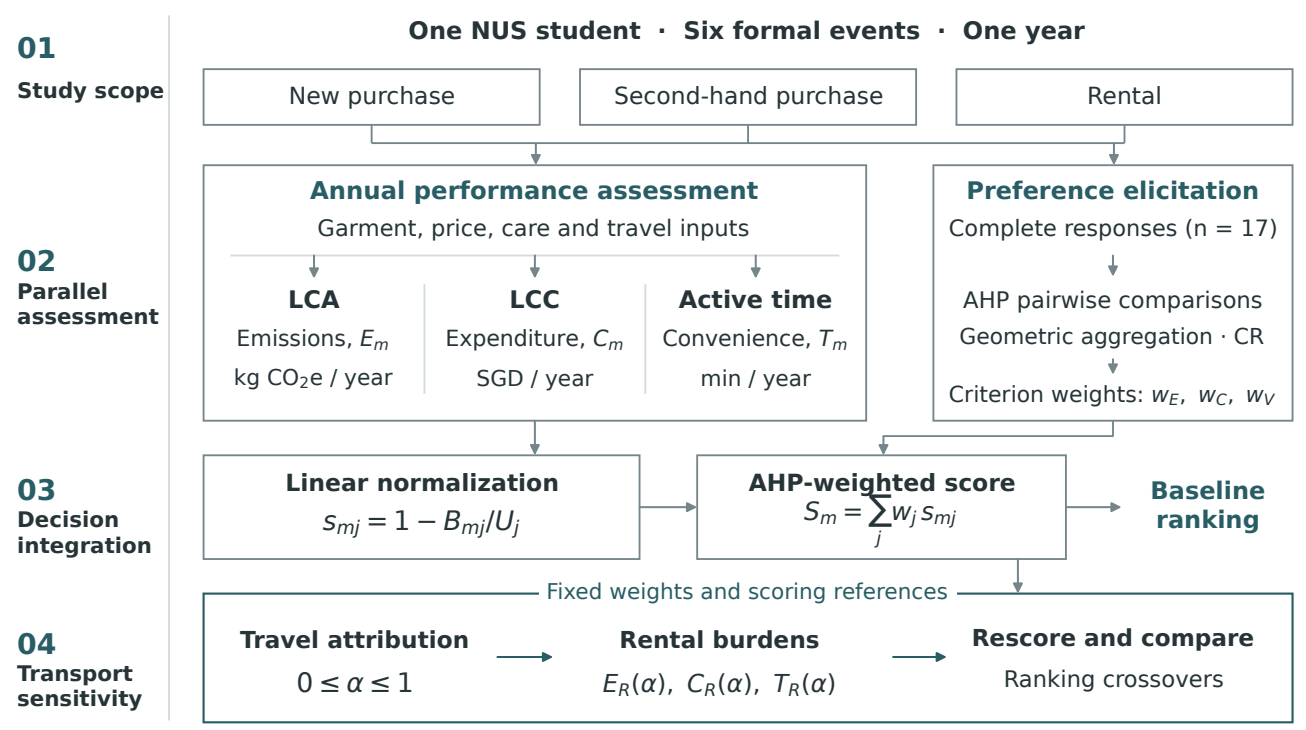
One NUS student · Six formal events · One year
01
Study scope
New purchase
Second-hand purchase
Rental
02
Parallel assessment
Annual performance assessment
Garment, price, care and travel inputs
LCA
Emissions, E_m
kg CO_2e / year
LCC
Expenditure, C_m
SGD / year
Active time
Convenience, T_m
min / year
Preference elicitation
Complete responses (n = 17)
AHP pairwise comparisons
Geometric aggregation · CR
Criterion weights: w_E, w_C, w_V
03
Decision integration
Linear normalization
$s_{mj} = 1 - B_{mj}/U_j$
AHP-weighted score
$S_m = \sum_j w_j s_{mj}$
Baseline ranking
Fixed weights and scoring references
04
Transport sensitivity
Travel attribution
$0 \le \alpha \le 1$
Rental burdens
$E_R(\alpha)$, $C_R(\alpha)$, $T_R(\alpha)$
Rescore and compare
Ranking crossovers